

PUTGARDEN KAP ARKONA, Germany

WMO: 100910

Lat: 54.6792N

Lon: 13.4342E

Elev: 42

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.1	-5.9	-11.9	1.4	-7.2	-9.8	1.6	-5.1	22.7	1.9	20.5	3.0	7.1	0	0.841

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	23.1	18.2	21.7	17.7	20.5	17.0	19.1	22.2	18.2	21.0	17.4	19.9	5.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	20.8	16.9	12.1	20.0	16.1	11.5	19.2	54.2	22.3	51.4	21.1	49.0	20.0	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.1	17.0	15.2	DB	-8.5	27.0	4.6	2.0	-11.9	28.4	-14.6	29.6	-17.1	30.8	-20.5	32.2	(4)
(5)			WB	-9.4	21.0	4.3	1.1	-12.4	21.8	-15.0	22.5	-17.4	23.1	-20.5	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	1.1	0.7	3.0	5.7	10.3	14.3	17.0	17.0	13.7	10.0	5.3	2.6
	DBStd	6.48	3.78	3.68	3.10	2.73	2.76	2.47	2.26	2.28	2.10	2.84	2.54	2.69
	HDD10.0	1322	275	262	219	132	30	1	0	0	1	34	141	229
	HDD18.3	3640	533	495	477	378	248	123	55	56	139	257	391	487
	CDD10.0	755	0	0	1	3	41	131	216	216	112	35	1	0
	CDD18.3	30	0	0	0	0	0	2	13	14	1	0	0	0
	CDH23.3	42	0	0	0	0	0	3	13	26	1	0	0	0
	CDH26.7	3	0	0	0	0	0	0	1	2	0	0	0	0
Wind	WSAvg	7.6	9.5	8.5	7.9	7.1	6.4	6.2	6.5	6.5	7.5	7.9	8.3	8.8
Precipitation	PrecAvg	528	34	29	32	37	44	40	56	62	51	46	56	44
	PrecMax	744	53	50	75	95	82	77	105	145	103	120	127	87
	PrecMin	333	9	6	9	3	13	13	10	9	12	5	19	16
	PrecStd	104	15	14	17	24	18	18	33	39	22	33	31	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	11.8	12.8	15.7	19.2	23.0	24.8	26.3	21.8	18.1	11.7	9.9
		MCWB	8.1	9.7	9.6	11.2	14.3	18.0	19.2	19.3	17.5	15.5	9.9	8.4
	2%	DB	7.3	8.6	9.6	13.0	16.9	21.1	23.1	23.3	19.3	15.7	10.3	8.7
		MCWB	6.2	6.7	7.7	8.9	13.1	17.2	18.3	18.4	16.2	13.5	9.0	7.7
	5%	DB	6.5	6.3	7.8	11.0	15.6	19.4	21.8	21.6	17.7	14.4	9.5	7.2
		MCWB	5.4	4.9	6.2	8.2	12.4	15.9	17.7	17.8	15.2	12.5	8.3	6.3
	10%	DB	5.7	5.0	6.6	9.4	14.4	18.1	20.5	20.4	16.7	13.5	8.7	6.3
		MCWB	4.7	3.8	5.3	7.3	11.6	14.9	17.0	17.0	14.3	11.7	7.6	5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	9.7	10.1	11.6	15.1	19.3	20.1	20.5	18.3	15.7	10.2	8.9
		MCDB	8.9	11.4	12.1	15.5	18.4	21.7	23.4	24.1	21.0	18.2	11.0	9.5
	2%	WB	6.4	6.6	7.8	9.6	13.7	17.6	18.9	19.2	16.7	13.8	9.2	7.6
		MCDB	7.1	8.2	9.3	12.1	16.3	20.2	22.1	22.4	18.6	15.2	10.0	8.6
	5%	WB	5.7	5.1	6.5	8.5	12.7	16.4	18.1	18.3	15.6	12.8	8.5	6.5
		MCDB	6.4	6.1	7.7	10.6	15.2	19.0	21.1	21.1	17.3	14.1	9.3	7.3
	10%	WB	4.7	3.9	5.3	7.5	11.8	15.2	17.3	17.4	14.5	12.0	7.7	5.5
		MCDB	5.6	4.9	6.5	9.2	14.1	17.8	20.1	19.8	16.4	13.2	8.6	6.2
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.3	3.7	4.6	4.8	4.9	4.9	5.1	4.2	3.7	3.3	3.3
		MCDBR	3.4	4.8	5.9	7.7	6.5	6.6	6.6	6.7	6.0	4.7	3.8	4.0
		MCWBR	3.3	4.0	4.5	5.0	4.5	4.4	4.3	3.9	4.2	3.8	3.6	3.8
	5% WB	MCDBR	3.3	4.7	5.7	7.0	6.0	6.3	6.1	6.0	5.6	4.3	3.7	3.8
		MCWBR	3.4	4.0	4.7	4.9	4.5	4.6	4.2	3.8	4.3	3.7	3.6	3.8
Clear-Sky Solar Irradiance	taub	0.319	0.341	0.367	0.396	0.392	0.387	0.408	0.409	0.377	0.362	0.341	0.311	
	taud	2.327	2.337	2.300	2.298	2.370	2.411	2.368	2.374	2.442	2.419	2.364	2.310	
	Ebn at Noon	618	737	800	827	853	861	833	808	788	706	584	539	
	Edh at Noon	56	78	100	114	112	109	112	105	87	72	55	47	
All-Sky Solar Radiation	RadAvg	0.47	1.08	2.39	4.16	5.33	5.72	5.29	4.28	3.06	1.55	0.62	0.33	
	RadStd	0.07	0.19	0.22	0.47	0.56	0.42	0.61	0.30	0.27	0.22	0.08	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-159	+60	N/A

Nomenclature: See separate page